

Foundations of Computer Vision

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(Many slides obtained from colleagues, and gratefully acknowledged)

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Scanning a document



What's written here?

We are unable to read the text in the white region clearly

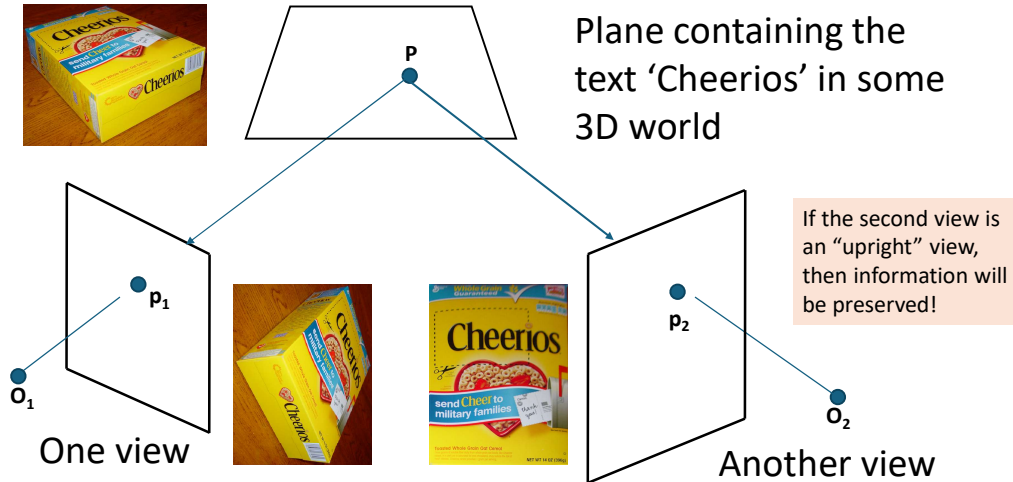
Concept:
Homography



Much better!

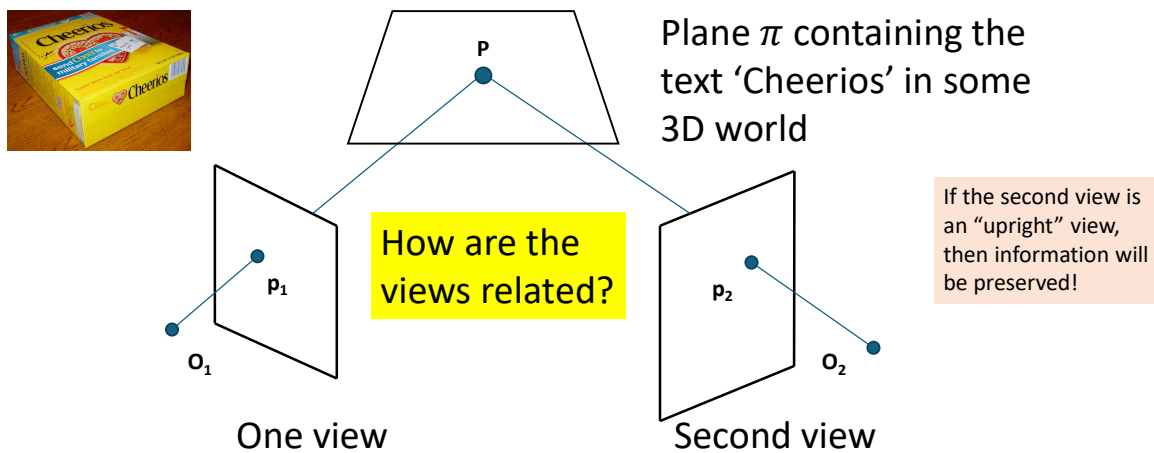
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Orthographic View



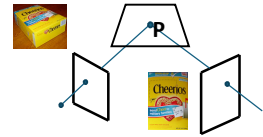
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Views



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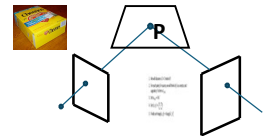
Recipe



- We will solve the scan problem using a recipe, and then answer why the method works
- Some parts of the recipe may appear weird
 - We will just accept these for now

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Recipe



1. We will discover a 3×3 matrix H
2. For each pixel p' in source, we will think of p' as a vector, and upgrade p' to form $u'_{3 \times 1}$
3. Set $u_{3 \times 1} = Hu'$
4. Set $(x, y) = (\frac{u_1}{u_3}, \frac{u_2}{u_3})$
5. Finally set $\text{Image}[x, y] = \text{Image}[x', y']$

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Loop over all pixels p' in source image
 $p \leftrightarrow p'$

Remove Distortion

Diagram illustrating the removal of distortion from a distorted image. The distorted image is shown on the left, with points A, B, and C marked. The undistorted image is shown on the right, with points A, B, and C marked. The transformation is defined by the equation:

$$D u = H \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix}$$

where D is a 3×3 matrix, u is a 3×1 vector, and H is a 3×3 matrix. The components are defined as:

$$x_1 = u_1 / u_3 \quad x_2 = u_2 / u_3$$

The relationship between the distorted and undistorted coordinates is given by:

$$I(x_1, x_2) = I(x', y')$$

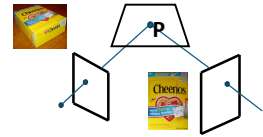
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How to discover H ?

1. The matrix H will appear as a solution to the equation $Ah = 0$ where h is the linearized for H
2. Our remaining goal is to set up the matrix A
3. Details
 1. To solve $Ah = 0$, we do the SVD of $A = UDV^T$
 2. We focus on V . If the rank of A is r , then the last $n - r$ columns of V is the basis for the null space.
 3. We claim the A **will** have a null space and the dimension of the null space is 1. In this way, we recover h .

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Notes



- The pictures have been taken casually by a layman
 - Not a scientist in the laboratory
- The output images appear very quickly
 - Computationally efficient
- The resulting picture are obtained by two-dimensional manipulations
 - But they are not the *snapchat* like manipulations
 - They seem to preserve the 3D appearance

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Recipe

- The recipe shown is elegant but not the one you should employ in the real world
 - Many important details missing
- Don't forget to revisit these slides after we reveal the magic (and thus the logic)

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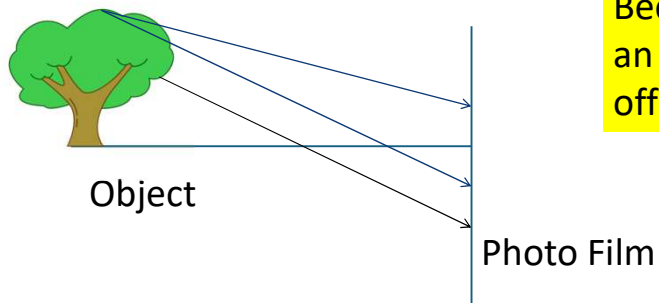
Agenda

- **Explanation** for the transformation equations
 - (x, y) doublet to triplet $u': (x, y, 1)$
 - The matrix equation $u = Hu'$

This relation is a homography, but not all such equations are homographies!

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Why do we see something?

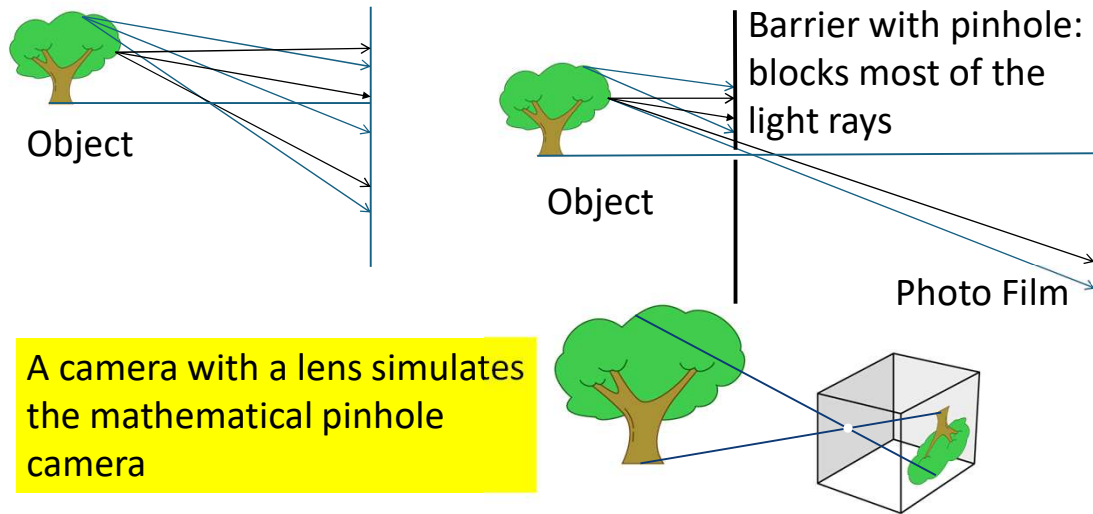


Because photons fall on an object, and bounce off and land in our eye

Will the eye see the object?

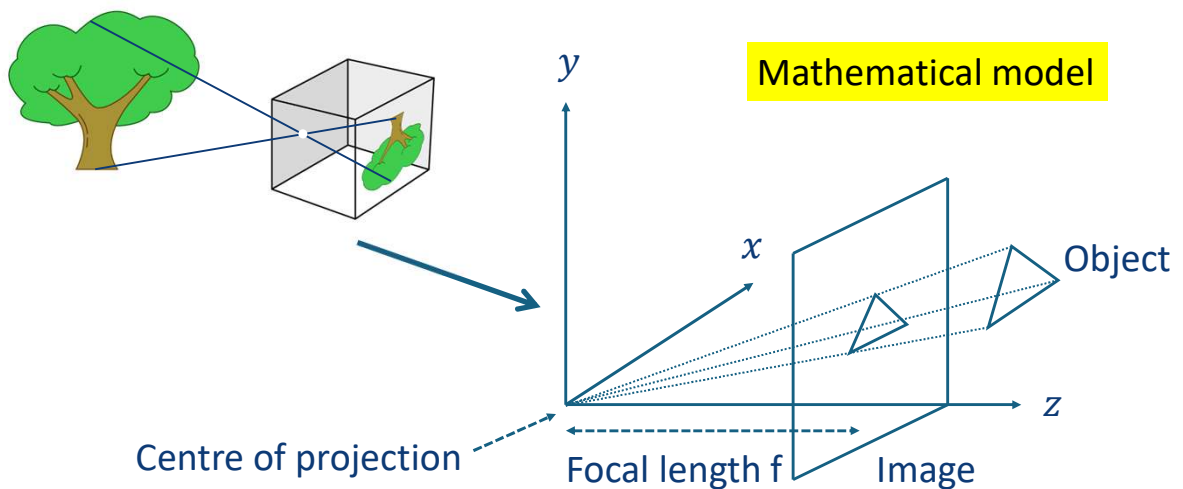
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Camera In Computer Vision



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Non-Inverted Image



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Image Formation

- The camera in computer vision is modeled in most cases as a pinhole camera
 - The optical lens make this possible

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Image Formation in a Pinhole Camera

- What is the relation between $\mathbf{p} = (x, y, z = f)$ and $\mathbf{P} = (X, Y, Z)$?
- In matrix form, we have:
 - non-linear, as the Z coordinate will vary from point to point.

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} f/Z & 0 & 0 \\ 0 & f/Z & 0 \end{pmatrix} \begin{pmatrix} X \\ Y \\ Z \end{pmatrix}$$

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Homogeneous Coordinates

- The relationship between (x, y) and (X, Y, Z) can be linearized using *homogeneous* coordinates on the lefthand side
 - And then moving to traditional Euclidean coordinates

$$\begin{pmatrix} x' \\ y' \\ w \end{pmatrix} = \begin{pmatrix} f & 0 & 0 \\ 0 & f & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} X \\ Y \\ Z \end{pmatrix}$$

$$x = \frac{x'}{w}, \quad y = \frac{y'}{w}$$

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Homogeneous Coordinates

- The relationship between (x, y) and (X, Y, Z) can be linearized using *homogeneous* coordinates on the left hand side
- We can also use homogeneous coordinates for the righthand side to write

$$x = P_{3 \times 4} X_{\text{cam}}$$

$$x \sim P_{3 \times 4} X_{\text{cam}}$$

$$\begin{pmatrix} x' \\ y' \\ w \end{pmatrix} = \begin{pmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} X \\ Y \\ Z \\ 1 \end{pmatrix}$$

**Important: This is not a real equality
and is a natural consequence of
Projective Geometry**

$$x = \frac{x'}{w}, \quad y = \frac{y'}{w}$$

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